

18. Two wires of the same material and length are connected in series in an electrical circuit. One wire is thicker than the other.

Which of the following statements is true when a current flows in the circuit?

- A The drift speed of electrons in the thicker wire is higher than the thinner wire.
- B The drift speed of electrons in the thicker wire is lower than the thinner wire.
- C The number density of charge carriers in the thicker wire is higher than the thinner wire.
- D The number density of charge carriers in the thicker wire is lower than the thinner wire.

Ans: B

- Since both wires are in series they must have the same current $I = nAvq$.
- If they are made of the same material, they will have the same number density of charge carriers n .
- The charge carriers (electrons) all have the same charge $q = e$.
- So the thicker wire (bigger A) will have the lower drift speed v .

19 A potentiometer circuit using a 1.0 m resistance wire of resistance $1.0\ \Omega$ placed